

STUDY QUESTIONS 2 ANSWER KEY

1. a) $\lambda_{eff} = \lambda_{in} + 0.20(\lambda_{eff}) \implies 0.8\lambda_{eff} = 30 \implies \lambda_{eff} = \frac{30}{0.8} = \mathbf{37.5}$ parts/hr.
 b) Total Capacity = $c \times \mu = 2 \times 25 = 50$ parts/hr.

$$\rho = \frac{\lambda_{eff}}{c\mu} = \frac{37.5}{50} = \mathbf{0.75} \text{ (or 75\%)}$$

2. Substitute $q = 100$ into the equation:

$$\begin{aligned} C(100) &= 10,000 + 5(100) + 500 \left(\frac{100}{200 - 100} \right) \\ &= 10,000 + 500 + 500 \left(\frac{100}{100} \right) \\ &= 10,500 + 500(1) \\ &= \mathbf{11,000} \end{aligned}$$

3. $\lambda = 30$, S1 (3 techs, 12/hr), S2 (2 eng, 20/hr), Rework 10%.

a) $\rho_1 = \frac{\lambda}{c\mu} = \frac{30}{3 \times 12} = \frac{30}{36} = \mathbf{0.833}$.

b) Effective flow into S2: $\lambda_2 = \frac{30}{1-0.1} = 33.33$. $\rho_2 = \frac{33.33}{2 \times 20} = \frac{33.33}{40} = \mathbf{0.833}$.

- c) Option 1 (Add to S1):

- S1 Cap: $4 \times 12 = 48$. S2 Cap: $2 \times 20 = 40$.
- $\rho_1 = 38/48 = 0.79$.
- $\rho_2 = 47.5/40 = 1.19$ (**Unstable/Bottleneck**).
- Max flow is limited by S2. S2 processes 40 units/hr total.
- System Output (Good parts) = $40 \times (1 - 0.2) = \mathbf{32}$ units/hr.

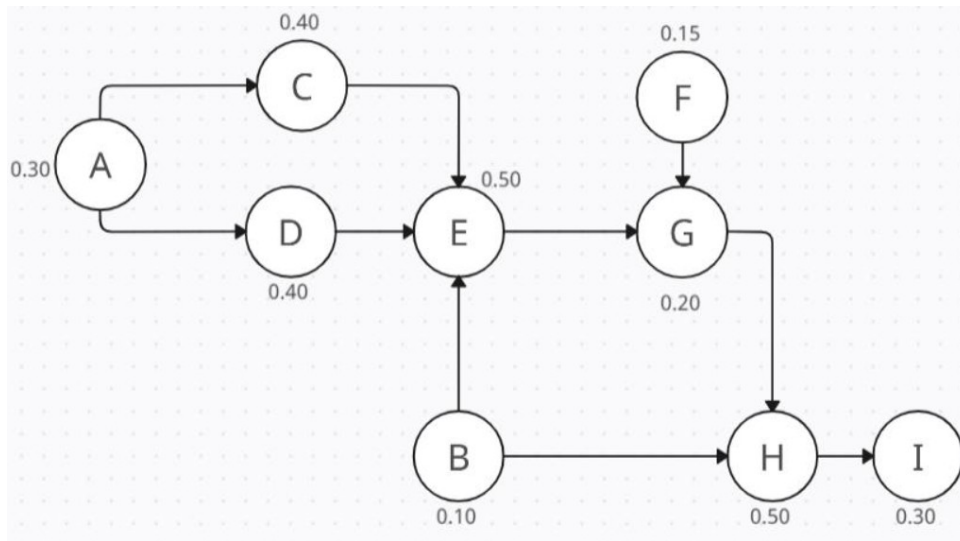
- d) Option 2 (Add to S2):

- S1 Cap: $3 \times 12 = 36$. S2 Cap: $3 \times 20 = 60$.
- $\rho_1 = 38/36 = 1.05$ (**Unstable/Bottleneck**).
- ρ_2 : Limited by S1 output. Flow to S2 is based on S1 max flow (36). $\rho_{2(actual)} = (36/0.8)/60 = 45/60 = 0.75$.
- Max flow limited by S1. S1 processes 36 units/hr.
- System Output (Good parts) = S1 Capacity = **36** units/hr.

- e) Comparison: Option 2 is better. It yields a production rate of 36 vs 32.

- Option 1 Bottleneck: Station 2 (Capacity 40 vs Demand 47.5).
- Option 2 Bottleneck: Station 1 (Capacity 36 vs Demand 38).

4. **Baking a Cake** Our precedence diagram is shown below:



For setting an assembly line, we need to determine the cycle time and minimum number of workstations:

- Total available time per week:

$$40 \text{ Hours} \times 60 \text{ Minutes per hour} = 2400 \text{ Minutes}$$

- Cycle time:

$$\frac{2400 \text{ minutes}}{2000 \text{ cakes}} = 1.2 \text{ minutes per cake}$$

- Minimum number of workstations:

$$N_{min} = \frac{\text{Total task time}}{\text{Cycle time}}$$

$$N_{min} = \frac{0.30 + 0.10 + 0.40 + 0.40 + 0.50 + 0.15 + 0.20 + 0.50 + 0.30}{1.2} = \frac{2.85}{1.2}$$

$$= 2.375 \approx 3 \text{ Workstations}$$

Let's prepare our assembly line:

Workstation	Time Remaining(min)	Eligible	Assigned Task	Idle Time(min)
1	1.20	A,B,F	A	0
	0.90	B,C,D,F	B	
	0.80	C,D,F	C	
	0.40	D,F	D	
	0	-	-	
2	1.20	E,F	E	0.35
	0.70	F	F	
	0.55	G	G	
	0.35	-	-	
3	1.20	H	H	0.40
	0.70	I	I	
	0.40	-	-	

Efficiency is given by:

$$\begin{aligned}
 Efficiency &= 1 - \frac{\text{Sum of Idle Times}}{\text{Number of Workstations} \times \text{Cycle Time}} \times 100 \\
 &= 1 - \frac{0.35 + 0.40}{3 \times 1.20} \times 100 \\
 &= 79,16\%
 \end{aligned}$$

5. **Automotive Factory Balancing** For setting an assembly line, we need to determine the cycle time and minimum number of workstations:

- Total available time per week:

$$50 \text{ Hours} \times 60 \text{ Minutes per Hour} = 3000 \text{ Minutes}$$

- Cycle Time:

$$\frac{3000 \text{ Minutes}}{130 \text{ Cars}} \approx 23 \text{ minutes per car}$$

- Minimum number of workstations:

$$N_{min} = \frac{\text{Total task time}}{\text{Cycle time}}$$

$$N_{min} = \frac{5 + 7 + 6 + 4 + 9 + 11 + 8 + 5 + 6 + 3 + 15}{23} = \frac{79}{23} = 3.43 \approx 4 \text{ workstations}$$

Part 1 – Longest Operation Time:

Workstation	Time Remaining (min)	Eligible	Assigned Task	Idle Time (min)
1	23	A,E	E	
	14	A,F	F	
	3	-	-	3
2	23	A	A	
	18	B,C	B	
	11	C	C	
	5	D	D	
	1	-	-	1
3	23	G	G	
	15	H,I,J	I	
	9	H,J	H	
	4	J	J	
	1	-	-	1
4	23	K	K	
	8	-	-	8

Part 2 – Shortest Operation Time:

Workstation	Time Remaining (min)	Eligible	Assigned Task	Idle Time (min)
1	23	A,E	A	1
	18	B,C,E	C	
	12	B,E	B	
	5	D,E	D	
	1	-	-	
2	23	E	E	3
	14	F	F	
	3	-	-	
3	23	G	G	1
	15	H,I,J	J	
	12	H,I	H	
	7	I	I	
	1	-	-	
4	23	K	K	8
	8	-	-	

Part 3 – Greatest Number of Following Tasks:

Workstation	Time Remaining (min)	Eligible	Assigned Task	Idle Time (min)
1	23	A,E	A	5
	18	B,C,E	B	
	11	C,E	C	
	5	-	-	
2	23	D,E	E	3
	14	D,F	F	
	3	-	-	
3	23	D	D	0
	19	G	G	
	11	H,I,J	I	
	5	H,J	H	
	0	-	-	
4	23	J	J	5
	20	K	K	
	5	-	-	

6. **Convert to a table** Our table which is converted from the precedence diagram given is shown below:

Element	Precedence	Time(hours)
A	-	0.3
B	A	0.6
C	-	0.5
D	B	0.8
E	C, D	0.9
F	-	0.5
G	E	0.4
H	F, G	0.1
I	G	0.7
J	H	0.6
K	I, J	1.2

7. **Work and Time Study** For determination of standard time, we need to compute the average observed time for each step separately:

$$\text{Average Observed Time} = \frac{\sum \text{Elemental Time (hours)}}{\text{Number of Cycles}}$$

For element 1:

$$\frac{9.54}{30} = 0.318 \text{ hours}$$

For element 2:

$$\frac{6.87}{30} = 0.229 \text{ hours}$$

For element 3:

$$\frac{16.45}{30} = 0.548 \text{ hours}$$

For element 4:

$$\frac{10.96}{30} = 0.365 \text{ hours}$$

Secondly, we calculate the normal time for each step:

$$\text{Normal Time} = \text{Average Observed Time} \times \text{Performance Rating}$$

For element 1:

$$0.318 \times 1.15 = 0.3657$$

For element 2:

$$0.229 \times 0.95 = 0.2176$$

For element 3:

$$0.548 \times 1.10 = 0.6028$$

For element 4:

$$0.365 \times 1.00 = 0.3650$$

Let's compute the total normal time:

$$0.3657 + 0.2176 + 0.6028 + 0.3650 = 1.5511 \text{ hours}$$

Finally, we can calculate the standard time:

$$\text{Standard Time} = \text{Normal Time} \times (1 + \text{Allowance Factor})$$

$$\text{Standard Time} = 1.5511 \times (1 + 0.09) = 1.5511 \times 1.09 = 1.6907 \text{ hours}$$

8. **Time Study of Valentine's Day** To compute the standard time for this job, first, we need to compute average observed time (We should not add outliers when we are doing our calculations):

$$\text{Average Observed Time} = \frac{\text{Total of elemental times}}{\text{Number of cycles}}$$

For element 1:

$$\frac{0.09 + 0.16 + 0.13 + 0.12 + 0.08 + 0.15 + 0.17}{7} = 0.129$$

For element 2:

$$\frac{0.05 + 0.15 + 0.14 + 0.08 + 0.09 + 0.07 + 0.11}{7} = 0.099$$

For element 3:

$$\frac{0.12 + 0.07 + 0.09 + 0.06 + 0.05 + 0.10 + 0.06 + 0.13}{8} = 0.085$$

For element 4:

$$\frac{0.06 + 0.07 + 0.11 + 0.08 + 0.08 + 0.07 + 0.07}{7} = 0.077$$

Secondly, we should compute the normal time:

$$\text{Normal Time} = \text{Average Observed Time} \times \text{Performance Rating}$$

For element 1:

$$0.129 \times 1.09 = 0.141$$

For element 2:

$$0.099 \times 1.06 = 0.105$$

For element 3:

$$0.085 \times 0.90 = 0.077$$

For element 4:

$$0.077 \times 1.10 = 0.085$$

Total normal time is:

$$0.141 + 0.105 + 0.077 + 0.085 = 0.408 \text{ minutes}$$

Let's calculate the standard time:

$$\text{Standard Time} = \text{Normal Time} \times (1 + \text{Allowance Factor})$$

Our allowance factor is 12% (or 0.12).

$$\text{Standard Time} = 0.408 \times (1 + 0.12) = 0.408 \times 1.12 = 0.457 \text{ minutes}$$

9. **Work and Time Study 2** For determining the production standard, standard labor cost and standard hours; we need to first calculate the standard time:

$$\text{Avg. Observed Time} = \frac{\sum \text{Elemental Time (min)}}{\text{Number of Cycles}} \text{ (We shouldn't add unrelated results)}$$

For element 1:

$$\frac{0.190 + 0.147 + 0.121 + 0.156}{4} = 0.1535 \text{ min}$$

For element 2:

$$\frac{0.033 + 0.033 + 0.014 + 0.046}{4} = 0.0315 \text{ min}$$

For element 3:

$$\frac{0.142 + 0.194 + 0.178 + 0.167}{4} = 0.1703 \text{ min}$$

For element 4:

$$\frac{0.062 + 0.088 + 0.060 + 0.056 + 0.075}{5} = 0.341 \text{ min}$$

Let's calculate the normal time:

$$\text{Normal Time} = \text{Avg. Observed Time} \times \text{Performance Rating}$$

For element 1:

$$0.1535 \times 0.95 = 0.1458$$

For element 2:

$$0.0315 \times 0.95 = 0.030$$

For element 3:

$$0.1703 \times 0.95 = 0.1618$$

For element 4:

$$0.341 \times 0.95 = 0.324$$

Total normal time:

$$0.1458 + 0.030 + 0.1618 + 0.324 = 0.6616$$

Standard time:

$$\text{Standard Time} = \text{Normal Time} \times (1 + \text{Allowance Factor})$$

$$\text{Total Allowance Factor} = 8\% + 4\% + 6\% = 18\%$$

$$\text{Standard Time} = 0.6616 \times 1.18 = 0.7807 \text{ minutes per piece}$$

Convert it to hours per piece:

$$\frac{0.7807}{60} = 0.013 \text{ hours per piece}$$

Let's compute the production standard (pieces per hour):

$$\text{Production Standard} = \frac{1}{\text{Standard Time(hours per piece)}}$$

$$\frac{1}{0.013} = 76.92 \approx 77 \text{ pieces per hour}$$

Secondly, compute the standard hours per 100 pieces:

$$\text{Standard Hours per 100 Pieces} = \text{Standard Time(hours per piece)} \times 100$$

$$0.013 \times 100 = 1.3 \text{ hours per 100 pieces}$$

Finally, compute the standard labor cost per 100 pieces:

$$\text{Standard Labor Cost per 100 Pieces} = \text{Std. Hours per 100 Pieces} \times \text{Wage per Hour}$$

$$1.3 \times 10 = 13 \text{ dollars per 100 pieces}$$

10. **Productivity** Partial productivity is calculated as:

$$\text{Partial Productivity} = \frac{\text{Output}}{\text{Input}}$$

a) Labor Productivity:

$$\text{Labor Productivity} = \frac{\text{Output}}{\text{Labor Hours}}$$

For the base year:

$$\frac{50}{265} = 0.189$$

For this year:

$$\frac{70}{310} = 0.226$$

b) Capital Productivity:

$$\text{Capital Productivity} = \frac{\text{Output}}{\text{Capital Investment}}$$

For the base year:

$$\frac{50}{12000} = 0.0041$$

For this year:

$$\frac{70}{19000} = 0.0036$$

c) Overhead Productivity:

$$\text{Overhead Productivity} = \frac{\text{Output}}{\text{Overhead Cost}}$$

For the base year:

$$\frac{50}{6500} = 0.0076$$

For this year:

$$\frac{70}{8000} = 0.00875$$

d) Machine Productivity:

$$\text{Machine Productivity} = \frac{\text{Output}}{\text{Machine Hours}}$$

For the base year:

$$\frac{50}{240} = 0.208$$

For this year:

$$\frac{70}{290} = 0.241$$

e) Total Productivity:

$$\text{Total Productivity} = \frac{\text{Total Output}}{\text{Total Input}}$$

For the base year:

$$\frac{50}{12000 + 6500 + 265 + 240} = \frac{50}{19005} = 0.00263$$

For this year:

$$\frac{70}{19000 + 8000 + 310 + 290} = \frac{70}{27600} = 0.00253$$